

CaseLinker: A Framework for Retrospective Analysis and Case Studies of Internet Crimes Against Children Across U.S. Task Forces, 2010–2026

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Independent Research | github.com/mrinaalr/CaseLinker | [Live Demo](#)

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Abstract

Histories of the internet are usually written from the perspective of platforms, growth, and innovation. Less documented is how the same infrastructure changed the conditions for child exploitation, and how law enforcement institutions adapted across successive technological shifts. Case-level evidence of that history exists in public press releases and task force reports, but the evidence is fragmented across jurisdictions and has not been connected at scale.

This report presents a framework for longitudinal case study analysis of the Internet Crimes Against Children (ICAC) Task Force Program from 2010 to 2026, organized into four technological eras: the social platform era (2010–2014), the mobile-first transition (2015–2018), platform proliferation (2019–2022), and the short-form social media and generative AI period (2023–2026). Drawing on 2,500 publicly available case reports processed through CaseLinker, the report specifies a methodology for 20 case studies that trace how offender methods, platform surfaces, and law enforcement responses changed across eras. Each case study is structured around five analytical sections: platform context, perpetrator methodology, investigative

approach, prosecutorial outcome, and relationship to its technological era. The framework, dataset, and analysis pipeline are released as open-source software so that investigators, researchers, and organizations in this domain can replicate and extend the work.

Keywords: ICAC investigations, retrospective case analysis, longitudinal study, internet history, child exploitation, open-source research infrastructure

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1. Introduction

1.1. A Fragmented Record

Histories of the internet’s first three decades document its platforms, companies, cultural shifts, and policy battles in detail. The parallel history of how offenders used the same technologies to reach and exploit children is far less complete.

Internet-facilitated crimes against children developed alongside the consumer web. Each wave of adoption reached offenders as it reached everyone else: chat rooms and peer-to-peer networks, then social media, mobile-first platforms, encrypted messaging, and generative AI. Law enforcement adapted in turn. The ICAC Task Force Program, created in 1998, now comprises 61 task forces and more than 6,200 federal, state, local, and Tribal agencies [14]. Investigative methods evolved over the same period, and prosecutorial outcomes accumulated in press releases and annual reports.

Prior research has measured parts of this history. The National Juvenile Online Victimization (N-JOV) studies surveyed national samples of law enforcement agencies in repeated waves, beginning with arrests made between July 2000 and June 2001 and continuing with 2006, 2009, and a fourth wave [27, 28, 10]. Those studies estimate the national volume and characteristics of arrests at fixed points in time. What remains unassembled is a continuous, case-level record, built from public documents, that links individual investigations across jurisdictions and technological eras. A companion essay sets out the motivation for this kind of retrospective analysis [19]. This report specifies how the CaseLinker corpus will be used to begin building that record.

1.2. The CaseLinker Series

This is the fourth report in the CaseLinker technical series. Report #1 introduced the system architecture: a modular five-layer pipeline for ingesting, processing, storing, clus-

tering, and visualizing publicly available ICAC case reports, evaluated on 47 cases from Arizona ICAC (AZICAC) annual reports covering 2011–2014 [16]. Report #2 scaled to 207 cases across additional sources, introduced named entity recognition via Stanford Stanza for agency extraction, and documented the distributed law enforcement network structure visible at that scale [17]. Report #3 scaled to 500 cases across five sources, introduced nine-dimensional facet tree navigation, and formalized the adversarial risk argument through a utility asymmetry proof [18].

This report does not introduce a new algorithmic component. Its contribution is the framework through which the CaseLinker corpus will support structured historical case studies: the methodology, sampling strategy, analytical sections, ethical and legal grounding, and execution plan.

CaseLinker Series Snapshot (April 2026)

Reports published: 4

Cases processed: 2,500+

Jurisdictions: 25 U.S. task forces and 1,000+ agencies

Features extracted: 27,150; 10 dimensions

HRPO Determination: #7668 (Not Human Subjects Research, 45 CFR 46)

1.3. Why Case Studies, Why Now

The first three reports surfaced patterns that appear only at scale: the distributed-core structure of ICAC agency networks, perpetrator age distributions, offense type composition, and jurisdiction-level variation in investigation types.

Aggregate statistics do not show how individual investigations unfolded. Aggregate counts report perpetrator ages and common platforms. A case study shows the sequence inside one investigation: which platform was involved, how contact was initiated, how investigators coordinated across agencies to build the case, and what the prosecution required to reach a conviction. Case-level detail also places each prosecutorial outcome within the evidentiary standards and priorities of its period.

The generative AI period adds urgency. NCMEC received 67,000 CyberTipline reports involving generative AI in 2024, up from 4,700 in 2023, a 1,325% increase; online enticement reports rose 192% to more than 546,000 over the same period [12]. Offender methods in the coming years are likely to differ from those of the past decade, and the

public record does not yet characterize the platforms and methods of recent cases in detail. A structured reading of past enforcement provides the baseline against which those changes can be measured.

At 2,500 cases, the CaseLinker corpus is large enough to support stratified case selection across all four eras.

2. Historical Framework: Four Technological Eras

Each of the four eras corresponds to a technological shift that changed both the conditions for exploitation and the law enforcement response. The eras are motivated by platform infrastructure and operationalized by the date of each case record. Transition dates are approximate, and cases near a boundary may share characteristics of the adjacent era.

2.1. Era 1: The Social Platform Era (2010–2014)

The early 2010s were the first period in which social media platforms reached mass adoption among adolescents. Facebook passed one billion users in 2012. Instagram launched in 2010 and was acquired by Facebook in 2012. Snapchat launched in 2011 with ephemeral messages as a core feature. These platforms changed the social environment of young people faster than parental awareness and legal frameworks adjusted.

In ICAC investigations, this era marks the emergence of platform-mediated contact as a primary grooming vector. CaseLinker’s original source, the 47-case AZICAC dataset covering 2011–2014 [16], reflects this transition: “online” and “chat” are the most frequent platform mentions (ten and seven cases), and Facebook appears in two cases, from 2012 and 2014. Investigation types in this subset range from reactive cases opened on parental reports and platform referrals to 12 proactive and 3 undercover operations. Counts at this scale describe the AZICAC subset and do not by themselves establish an era-wide trend.

2.2. Era 2: The Mobile-First Transition (2015–2018)

By early 2015, 73% of U.S. teens aged 13–17 had or had access to a smartphone [9]. Internet access moved from a shared family computer to a personal, always-connected device, and the constraints that had bounded contact (a child at a computer, at home, during limited hours) largely fell away.

The same period brought new contact surfaces that were harder to detect and document. A third of teen smartphone users already used messaging apps such as Kik and WhatsApp by 2015 [9]; anonymous-sharing apps such as Whisper and disappearing stories spread alongside them. In April 2016, WhatsApp made end-to-end encryption the default for all users [22]. ICAC case reports from this period describe greater forensic complexity: multiple devices, multiple platforms, and longer evidence chains.

2.3. Era 3: Platform Proliferation (2019–2022)

From 2019 to 2022 the platform environment fragmented and report volume accelerated. TikTok reached mass adoption in Western markets after its 2018 merger with Musical.ly. Discord, launched as a gaming communication tool, became a general-purpose platform heavily used by adolescents, and livestreaming services added real-time contact surfaces. The facet tree shows a corresponding increase in platform diversity among successful operations: 795 unique cases with social media platform involvement in this era, compared with 172 in Era 1 and 94 in Era 2.

COVID-19 school closures and social isolation moved more of children’s lives online, and Europol reported increased online child sexual abuse activity, including CSAM distribution, during the 2020 lockdowns [4]. Across the full period of this study, CyberTipline volume rose from roughly 100,000 reports in 2010 to 36.2 million in 2023 [26]. Because hash-based detection generates the large majority of industry reports [25], that growth reflects expanded platform detection as well as underlying offending.

2.4. Era 4: Short-Form Social Media and Generative AI (2023–2026)

The current era combines two developments: short-form social media as the dominant adolescent platform environment, and generative AI as both a threat vector and a detection problem. NCMEC received 67,000 CyberTipline reports involving generative AI in 2024, up from 4,700 in 2023 [12]; NCMEC notes that not every AI-related report involves AI-generated CSAM [11]. The Internet Watch Foundation’s July 2024 update documented the first realistic AI-generated videos of child sexual abuse [7], and Australia’s eSafety Commissioner recorded a 218% increase in reports of AI-generated abuse material between 2023 and 2024 [3]. Synthetic imagery requires investigators to determine whether material depicts a real child before committing victim-identification resources [3, 6]. Newly generated images also fall outside hash matching, which detects only previously verified material [20], and research has documented limited robustness to cropping and practical evasion attacks against PhotoDNA, the most widely deployed perceptual

hash for this purpose [24, 2].

Financial sextortion emerged as a distinct and fast-growing crime typology: NCMEC received nearly 100 financial sextortion reports per day in 2024, within a total of more than 546,000 online enticement reports [12, 13]. The CaseLinker corpus contains 1,147 case reports from this era describing completed operations.

The law enforcement response in this era draws on an ICAC program that dates to 1998 [14] while absorbing this volume. The REPORT Act of 2024 extended mandatory reporting to online enticement and child sex trafficking [13]. NCMEC testimony describes declining reports from several historically strong reporters and anticipated reporting losses as platforms adopt end-to-end encryption [23].

Table 1: Four-Era Framework Summary

Era	Period	Defining Technology	LE Context
Social Platform	2010–2014	Facebook, Instagram, Snapchat reach mass adolescent adoption	Platform-mediated contact emerges as a grooming vector; reactive, proactive, and undercover investigations
Mobile-First	2015–2018	Majority teen smartphone access (73% by 2015); messaging and anonymous apps; default E2EE on WhatsApp (2016)	Multi-device, multi-platform evidence chains
Platform Proliferation	2019–2022	TikTok, Discord, livestreaming; COVID-19 shift to online life	Case volume and platform diversity increase
Short-Form Social Media and Generative AI	2023–2026	AI-generated CSAM; financial sextortion; E2EE expansion	Novel imagery outside known-hash detection; REPORT Act (2024); new crime typologies

3. Research Methodology

3.1. Grounded in CaseLinker

This research is grounded in CaseLinker, an open-source retrospective case analysis system that processes, structures, and surfaces patterns in publicly available ICAC press

releases and task force case reports. The system is auditable and interpretable, operates only on public, already-redacted case summaries, and produces no outputs that identify individuals. The source corpus, extraction code, and analysis pipeline are released as open-source software to support replication and extension.

At the time of this report, CaseLinker has processed 2,500+ publicly available case reports across 25 U.S. jurisdictions, extracting features across ten dimensions: topic, severity, platform, investigation type, source, agency, prosecution outcome, location, severity phrase, and date range. The 20 case studies are selected from this corpus using the procedure below.

3.2. Case Study Selection: Stratified Purposive Sampling

The 20 case studies are selected by stratified purposive sampling. Cases are stratified by era and chosen within each era to span variation on five stratification variables, a diverse-case design in the sense of Seawright and Gerring [21]. The aim is analytic illustration, not statistical generalization: a small set of well-chosen cases can show how patterns and institutional responses played out in practice, which neither aggregate statistics nor unstructured reading of individual cases provides [5].

Era distribution: Cases are sampled across all four eras in approximate proportion to the corpus coverage of each period, with a minimum of four cases per era so that every era is represented.

Stratification variables: Within each era, cases are selected to represent variation across:

1. **Platform involvement:** from early social media contact to encrypted messaging to recent online sextortion
2. **Victim and perpetrator demographics:** age, relationship, and relevant context
3. **Investigation type:** proactive, reactive, and undercover operations
4. **Geography and jurisdiction:** variation across task forces, states, and agency networks
5. **Agency coordination:** single-agency cases versus multi-agency and large-scale operations

Candidate cases are identified with CaseLinker’s ten-dimensional facet tree, which filters by era (derived from date range), platform, investigation type, severity, and source. The facet tree organizes the 2,500-case corpus into a navigable structure of 12,402+ nodes. Selection proceeds in three steps: facet pruning identifies cases matching a target stratum; candidates are ranked by feature completeness and by how well they represent the era; and each candidate is checked against the public records pipeline to confirm whether supplementary redacted summaries are available. Ranking by feature completeness favors agencies that publish detailed releases, a selection effect recorded in Section 5.4.

3.3. Case Study Structure

Each case study is written as a structured narrative across five analytical sections:

1. **Platform context:** the digital environment in which the offense occurred, which platforms were used and how, and what that use reflects about the technological period
2. **Perpetrator methodology:** the behavioral pattern of the offense, including whether and how contact was initiated, how grooming or abuse unfolded where applicable, any connection to possession of child sexual abuse material or other offenses, and what the offense type reveals about the era
3. **Investigative approach:** how law enforcement identified, investigated, and built the case, including investigation type, agency network, and digital forensic methods
4. **Prosecutorial outcome:** charges, conviction, and sentencing, examined for what they reveal about the evidentiary standards and prosecutorial priorities of the period
5. **Era relationship:** how the case reflects, complicates, or extends the patterns of its technological era, and how the case connects to the other 19 case studies

3.4. Source Materials and Legal Grounding

All case studies draw only on public, already-redacted records. The primary source corpus is the 2,500-case CaseLinker dataset, drawn from publicly available ICAC press releases and annual task force reports. Where supplementary redacted summaries add analytical depth, records will be requested under the applicable state open records statutes:

- Arizona: Public Records Law (A.R.S. § 39-121 et seq.)
- Georgia: Open Records Act (O.C.G.A. § 50-18-70 et seq.)

- Florida: Public Records Law (Fla. Stat. ch. 119)
- Texas: Public Information Act (Tex. Gov’t Code ch. 552)
- California: California Public Records Act (Cal. Gov’t Code § 7920.000 et seq.)
- New York: Freedom of Information Law (N.Y. Pub. Off. Law §§ 84–90)
- Illinois: Freedom of Information Act (5 ILCS 140)

All records requests are limited to closed, adjudicated cases to reduce exemption risk under state law. No attempt will be made to re-identify individuals, reconstruct victim or perpetrator identity beyond what appears in published records, or supplement public documents with external investigation.

The University of Massachusetts Amherst Human Research Protection Office determined (HRPO Determination #7668, March 10, 2026) that this research is not human subjects research: the project does not involve intervention or interaction with individuals or use identifiable private information [45 CFR 46.102(f)(1), (2)].

4. CaseLinker as Research Infrastructure

4.1. From Tool to Infrastructure

The first three reports documented CaseLinker as a system: its architecture, extraction methodology, clustering and triage logic, and analytical outputs. For case study work, CaseLinker serves as research infrastructure: the case studies depend on a corpus that has already been collected, structured, and made queryable.

Without the facet tree, selecting 20 cases across four eras and five stratification variables would require reading all 2,500 case summaries. Studies of investigators who view child sexual abuse material report secondary traumatic stress and burnout, with greater exposure associated with higher symptom levels [15], and document the coping demands of this work [1]. Those studies measure exposure to images and video. The CaseLinker corpus consists of redacted text summaries, a lower-intensity exposure that the literature has not measured. Reducing cumulative exposure remains a design goal, consistent with law enforcement tooling efforts that aim to cut both analysis time and the emotional toll of review [26]. The facet tree turns candidate identification into a sequence of queries and

limits full-text reading to cases already matched to a stratum.

The case studies are therefore selected from a processed corpus rather than assembled ad hoc. CaseLinker has extracted domain features from 2,500 cases and can surface cases that represent each era and stratification variable, including cases that unstructured manual search would be unlikely to find.

4.2. Extensibility and Replication

CaseLinker was designed for extensibility from its initial release: the architecture is modular, the dataset is open, and the case study framework established here is replicable.

The 2,500-case corpus and the extraction pipeline are released as open-source software. An investigator at an ICAC task force, a researcher at NCMEC, or a journalist covering technology-facilitated harm can apply the same framework to cases outside the 20 studied here without rebuilding the pipeline. The sampling procedure, the five-section case structure, and the records request framework are documented in this report for that purpose. The intended output is a replicable method and an open corpus that others can extend.

4.3. Vicarious Trauma and Domain Considerations

Research in this domain carries psychological costs for everyone who engages with it closely, researchers included. The case studies therefore present structured narratives across the five analytical sections without reproducing graphic case details, dwelling on offense specifics beyond what the analysis requires, or presenting material designed to provoke outrage rather than understanding.

The same restraint applies to how offenders are characterized. Moral labels are common in public discussion of these crimes, but labels do not explain how offending changed. Understanding how offenders adapted to platform changes, including how some found communities on new platforms, and how law enforcement responded requires treating cases as evidence about a systemic problem. The unit of analysis is the pattern across cases (the technological period, the enforcement response, and the implications for prevention), not the individual offender.

5. Ethical and Dissemination Framework

5.1. Research Ethics and HRPO Determination

The ethical grounding of this research rests on three principles applied across the CaseLinker series: exclusive use of public, already-redacted records; no attempt to re-identify individuals; and review before any public release.

HRPO Determination #7668 from the University of Massachusetts Amherst Human Research Protection Office establishes that this research does not involve intervention or interaction with individuals or use identifiable private information [45 CFR 46.102(f)(1), (2)]. The determination reflects the nature of the source material, case records that law enforcement agencies published for public release. The determination is not a judgment that the subject matter lacks sensitivity; that sensitivity is the reason for the safeguards documented here.

No case study includes information that risks identifying victims, witnesses, or individuals not named in the original published records. Convicted offenders' names appear in court records and press releases but are omitted from the case studies. Cases with the highest severity indicators, or with details that could compromise ongoing investigations or reveal non-public law enforcement methods, are further abstracted or withheld from public release.

5.2. Restricted-First Dissemination

Case study findings are shared with practitioner and research contacts before public release. Pre-release review is not legally required, since the records are public and the research holds a not-human-subjects determination. The decision is professional: investigators, task force commanders, and child safety researchers are better placed than any automated system to judge whether a case study derived from public records carries risks that aggregate review would miss.

The practitioners, collaborators, and mentors who have engaged with CaseLinker have helped validate its utility and scope, and restricted-first dissemination keeps them in the review loop.

5.3. Goal of This Work

Twenty case studies cannot capture the full history of internet-facilitated crimes against children. Their purpose is to make the operational reality of these cases legible in ways that aggregate statistics do not.

Officers and commanders in this work build cases from digital fragments, coordinate across agencies, and make triage decisions under time pressure and uncertainty, the conditions studied in naturalistic decision-making research [8]. That institutional knowledge has not been systematically documented. Technology companies have also contributed through hash-based detection, shared hash databases that let one platform detect material first verified on another, and cooperation with law enforcement [25, 20]. Twenty structured case studies across four eras can show what that work looked like in practice: which investigations succeeded, how methods evolved, and how agency networks formed.

The case studies also lower the barrier to entry for people outside the field. A researcher who has never read an ICAC case report, a policy analyst who has never spoken with a task force commander, or a legislator who has not seen what an investigation involves can learn more from a structured case than from a platform involvement percentage. A case study from 2013 showing how initial contact occurred through early Facebook, how investigators built the case, and how the prosecution succeeded gives these readers an entry point into a domain that is otherwise opaque. The aim is more people working on this problem with better information.

5.4. What This Research Does Not Do

Overstatement carries real consequences in this domain, so the limits of the framework are stated explicitly. The case study framework does not:

- Re-identify or further identify victims, offenders, or witnesses beyond what appears in source publications
- Reconstruct operational law enforcement methodology beyond the source material
- Make statistical generalizations beyond what the corpus and sampling design support
- Treat cross-era differences in case counts as population trends: source coverage varies by era, so counts partly reflect corpus composition
- Claim neutral case selection: ranking by feature completeness favors agencies that publish detailed releases
- Supplement public records with external investigation, commercial data, or non-public sources
- Claim to represent the full population of ICAC investigations: the corpus covers pub-

licly reported cases, and the case studies cover only the closed, adjudicated subset

The adversarial risk argument formalized in Report #3 applies here. Structured case studies of successful prosecutions add no failure modes, operational tradecraft, or evasion-relevant information beyond what manual review of the source documents already provides, so the utility asymmetry between defenders and adversaries is unchanged.

6. Roadmap and Next Steps

6.1. Case Study Execution Timeline

This report establishes the framework. Case studies are scheduled for summer 2026, following the methodology in Section 3. The execution sequence is:

1. Finalize stratified case selection using CaseLinker facet tree navigation across all four eras and five stratification variables
2. Submit open records requests for supplementary redacted summaries under applicable state statutes, limited to closed, adjudicated cases
3. Draft case studies across the five analytical sections
4. Conduct practitioner review before public release
5. Release publicly as Report #5 or as a standalone case study document, depending on the volume and scope of completed studies

6.2. Parallel CaseLinker Development

The case study work runs in parallel with continued CaseLinker development. Planned additions for this period include:

- **Triage scoring refinement:** extending the priority triage system to surface cases relevant to specific investigative contexts, not only severity
- **Temporal trend analysis:** structured tracking of platform involvement, offense type composition, and investigation type distribution across the four eras
- **Cross-era pattern detection:** identifying cases that bridge eras or show methods characteristic of transitional periods

- **Expanded geographic coverage:** additional task force sources to increase jurisdictional diversity and reduce single-source reporting bias

These additions serve the case study work directly: richer analytical infrastructure supports better-grounded case selection and more defensible pattern claims.

6.3. Community Engagement and Extension

The framework and corpus are open. Researchers, journalists, ICAC practitioners, and child safety organizations are encouraged to apply the methodology to additional cases, extend the corpus, or pursue their own questions with the existing dataset. The records request framework, the extraction pipeline, and the five-section case structure are documented and replicable, so the next researcher does not need to rebuild them.

7. Conclusion

The history of internet-facilitated crimes against children since 2010 is recorded in public case reports, more than 2,500 of which are now in the CaseLinker corpus. Those reports sit in annual reports and press releases across jurisdictions and had not been connected at scale. This report provides a framework for reading them systematically and for turning that reading into grounded historical case studies that practitioners, researchers, journalists, and the public can use.

The four eras reflect technological shifts visible both in the corpus and in external measures of platform adoption and reporting volume across 16 years of public ICAC records. The 20 case studies trace how offenders adapted, how platforms became contact vectors, how task forces responded, and what the prosecutorial record shows when cases are read across eras rather than in isolation.

Every methodological choice documented here and in earlier reports (the stratified purposive sampling procedure, the five-section case structure, the records request framework, and the ethical safeguards) is designed for replication. Protecting children from online harm is too large a problem for any single system or researcher, and open infrastructure lets more people study this history rigorously without unnecessary exposure to its worst material. The subject matter is serious and the cases are real, which is why the framework requires methodological care and practitioner review at each step.

The case studies built on this framework will be released as Report #5.

CaseLinker is available at github.com/mrinaalr/CaseLinker under the MIT License.
Live demo: web-production-13a2.up.railway.app.

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